ARIZONA STATE UNIVERSITY

CSE 434 — COMPUTER NETWORKS — FALL 2026

SOCKET PROGRAMMING PROJECT
FINAL PROJECT DESIGN REPORT

Distributed Hash Table with Ring Management and Hot-Potato Query Processing

Item	Details
Group	33
Leader peer	falabbad
Language/transport	Python 3 / UDP
Ports	17500–17999
Full demo	6 peers; n=5; year 1996
Supplied data	1950: 223 rows, 51 columns, s=449

1. Executive Summary

The program provides the ten manager commands in the project specification, peer-to-peer ring construction, record storage, hot-potato querying and churn/teardown workflows. The supplied 1950 dataset contains 223 rows and 51 columns; its hash table size is 449. The full-project demo requires details-1996.csv, which must be downloaded using the included script.

2. Manager commands

Command	Purpose
register	Register peer as Free.
setup-dht	Select n members and set leader.
dht-complete	Confirm construction.
query-dht	Free peer gets a random DHT start node.
leave-dht	Tear down, renumber, rebuild n–1 members.
join-dht	Rebuild to include Free peer.
dht-rebuilt	Confirm pending churn operation.
deregister	Remove Free peer.
teardown-dht	Leader initiates deletion.
teardown-complete	All former members become Free.

3. Architecture, ports and hashing

Peers have separate manager and P2P UDP sockets. Group 33 ports must be 17500–17999. Hashing is position=EVENT_ID mod s and destination=position mod ring_size. Local hash positions map to lists, so collisions do not overwrite records.

Process	m-port	p-port
Manager	17500	—
falabbad	17501	17502
peerone	17503	17504
peertwo	17505	17506
peerthree	17507	17508
peerfour	17509	17510
peerfive	17511	17512

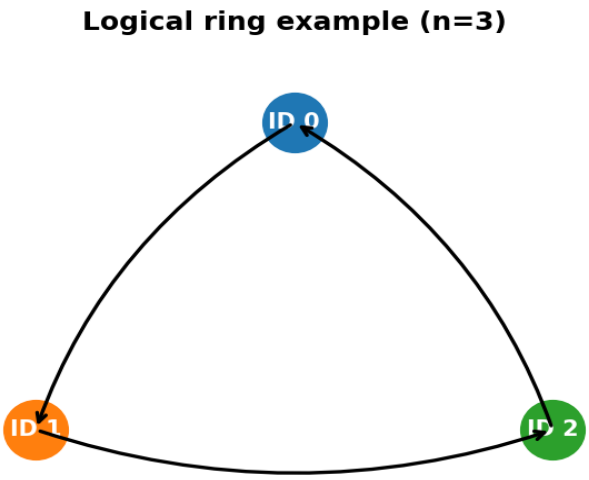


Figure 1. Logical ring example.

4. Wire protocol

JSON messages use named fields. Key P2P types are set-id, set-id-ack, store, store-ack, store-complete, count-report, find-event, find-event-response, teardown, reset-id, rebuild-dht, join-request and rebuild-complete.

```
{"command":"register","peer_name":"falabbad","ip":"192.0.2.10","m_port":17501,"p_port":17502}
{"command":"setup-dht","peer_name":"falabbad","n":5,"year":1996}
```

5. Time-space diagrams

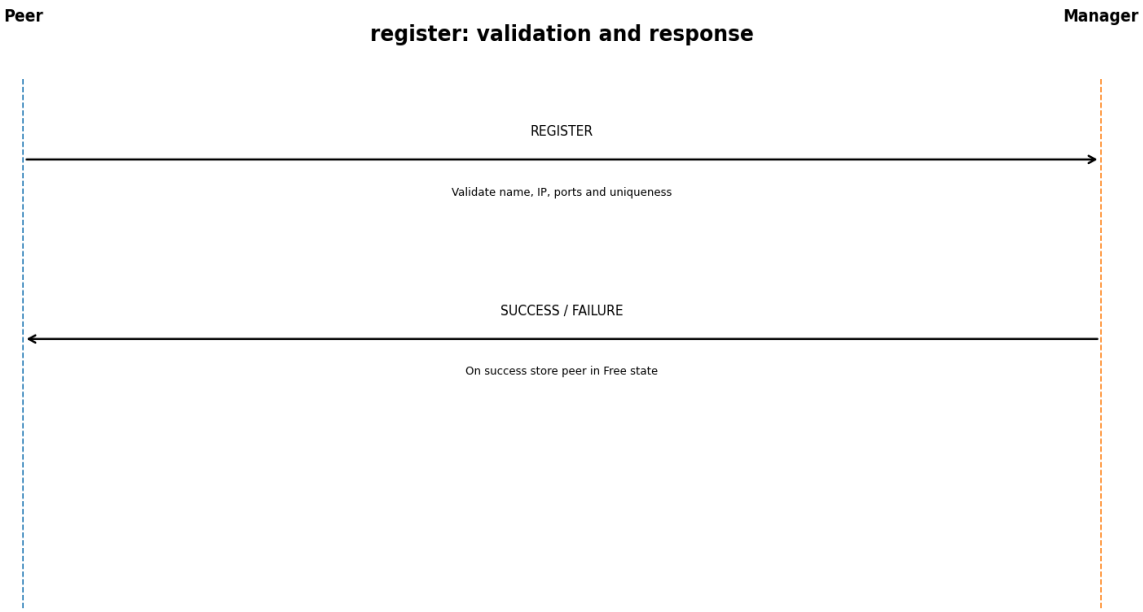


Figure 2. register.

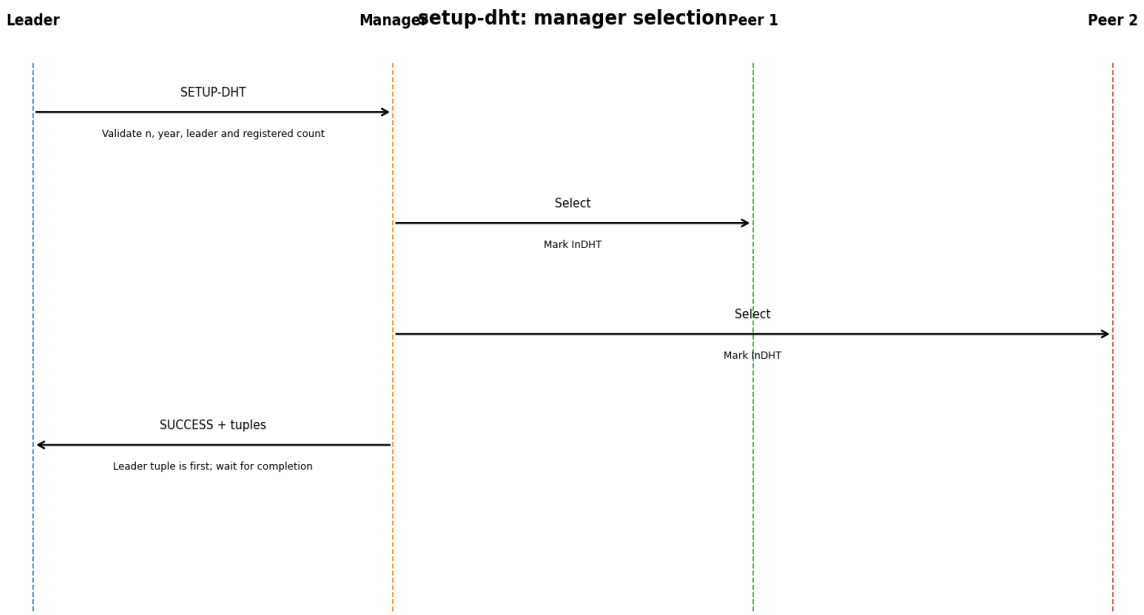


Figure 3. setup-dht.

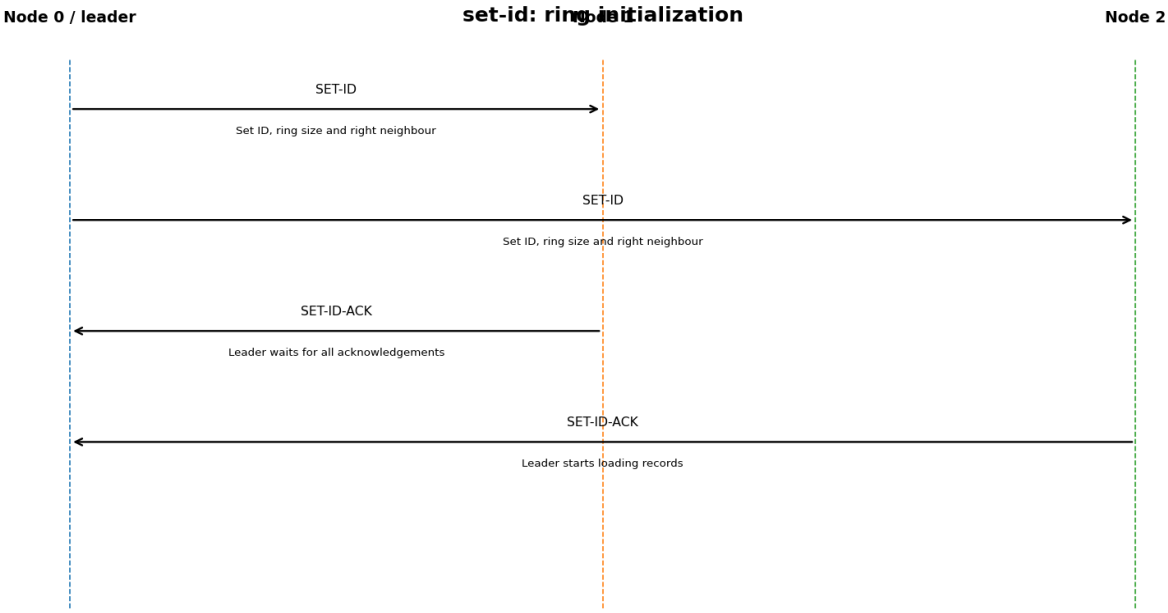


Figure 4. set-id.

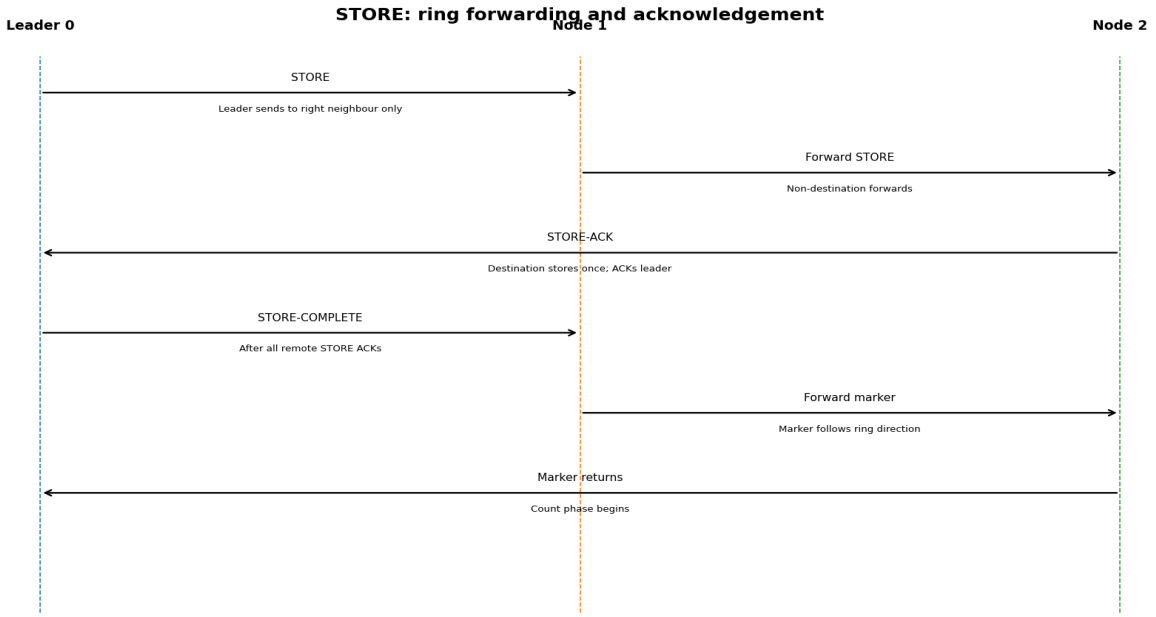


Figure 5. STORE and acknowledgement.

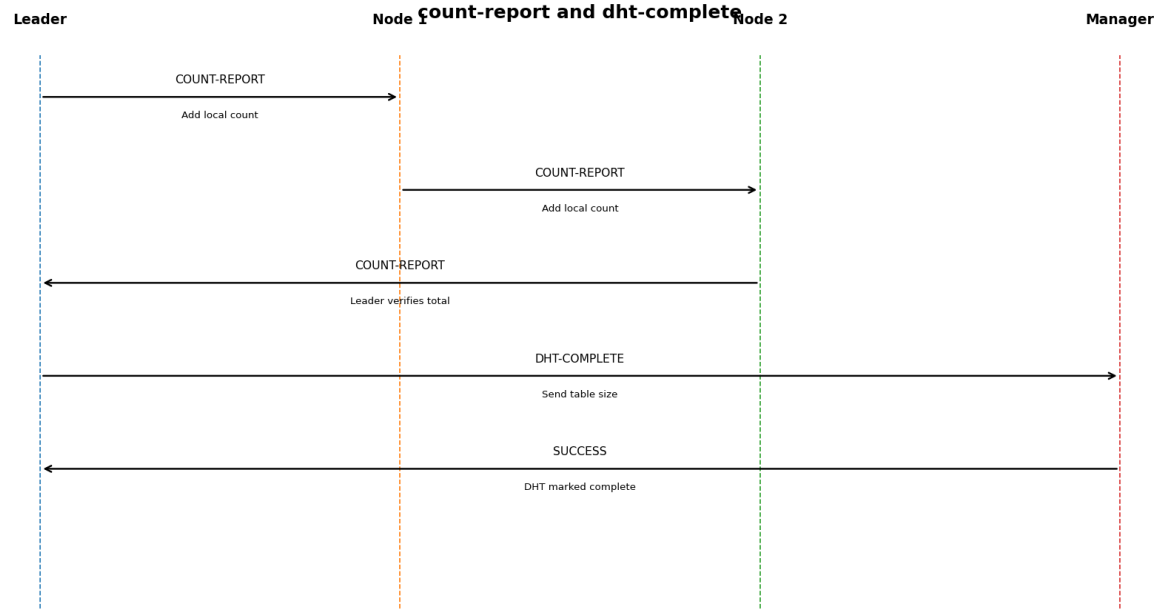


Figure 6. count-report and dht-complete.

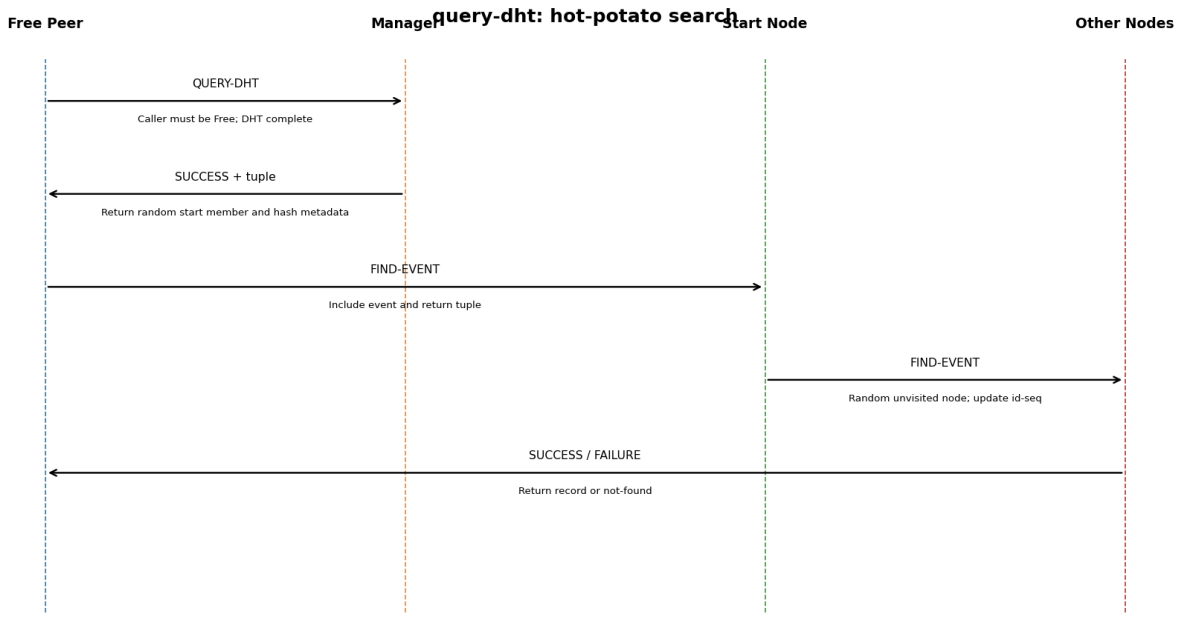


Figure 7. query-dht hot-potato.

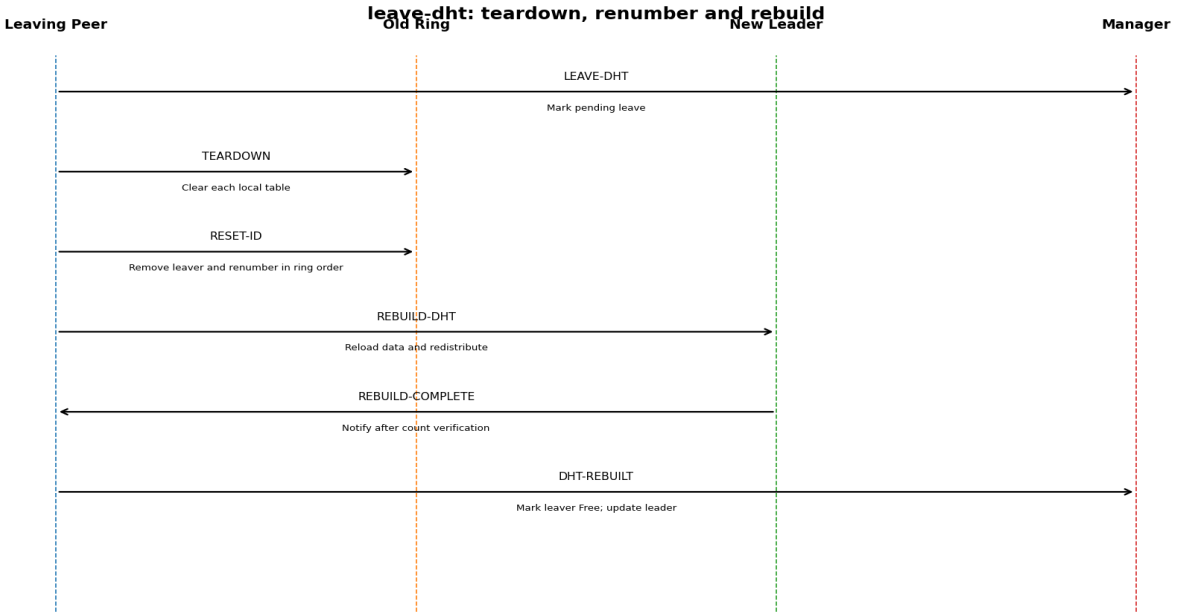


Figure 8. leave-dht rebuild.

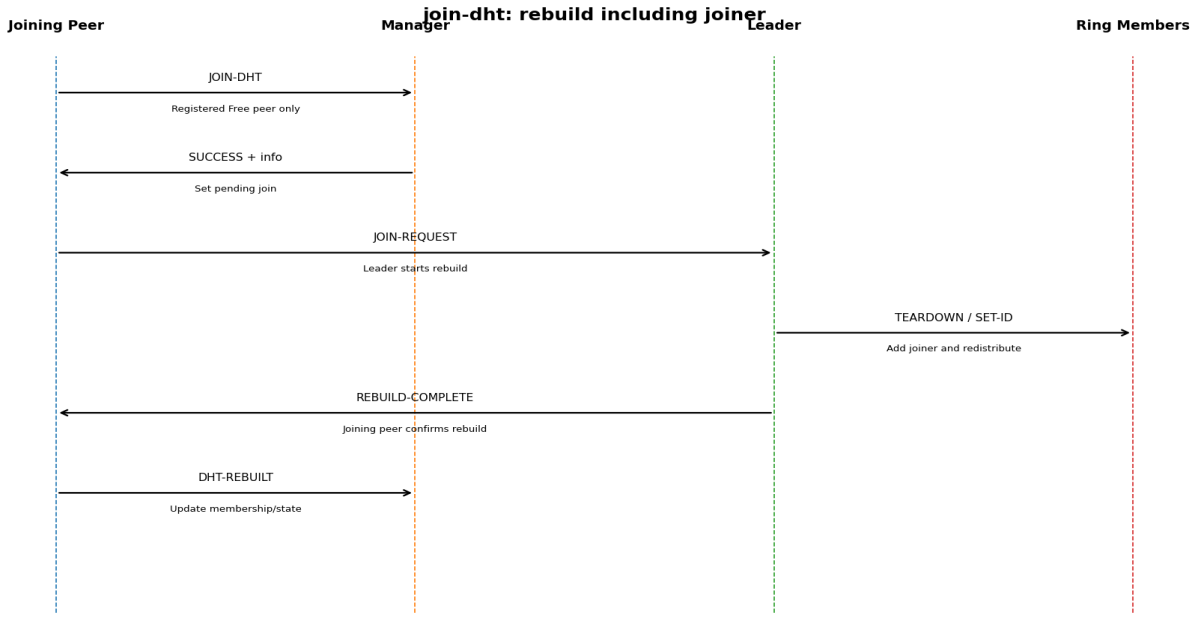


Figure 9. join-dht rebuild.

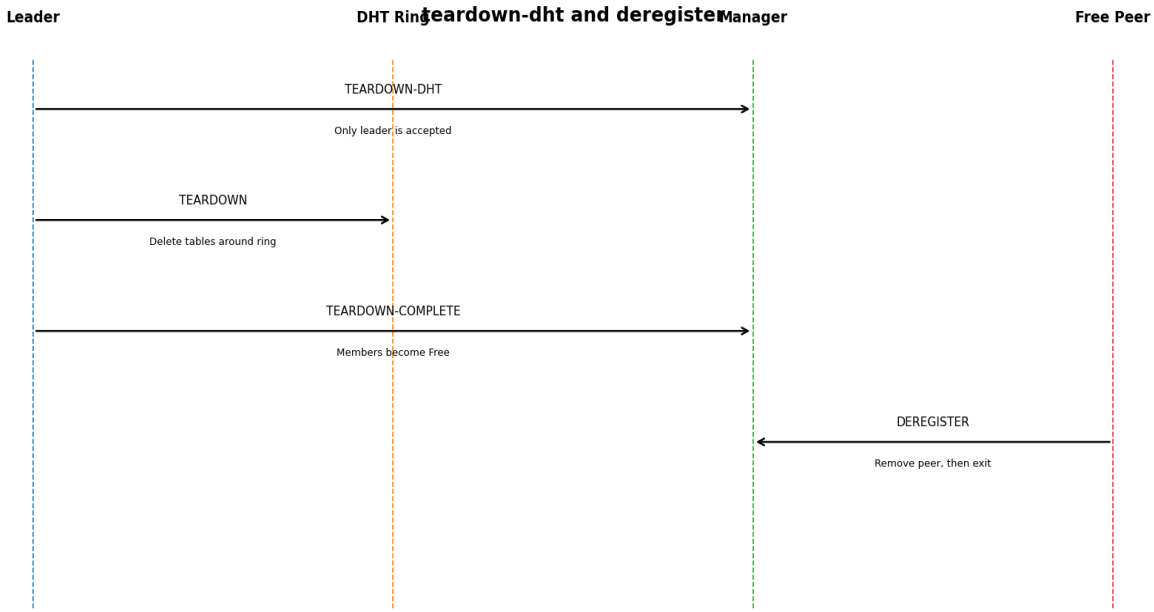


Figure 10. teardown and deregister.

6. Run instructions

```
python3 -m py_compile manager.py peer.py protocol.py dht_utils.py download_dataset.py
python3 manager.py 17500
python3 peer.py 17500
register falabbad 17501 17502
python3 download_dataset.py 1996
setup-dht falabbad 5 1996
query-dht 5536849
leave-dht
join-dht
teardown-dht
dereg
```

Use actual reachable IPv4 addresses; do not use loopback across hosts. The full demo needs at least four end-hosts and six peer processes.

7. Verification and submission items

Check	Status
Syntax check	Passed locally.
3-peer 1950 DHT setup and 223-record count	Passed locally; s=449.
dht-complete	Manager SUCCESS observed.
Multi-host full-project demo with 1996, queries, join, leave and teardown	Must be verified on actual course hosts.
Commit screenshot, video URL and timestamps	Must be inserted from actual repository/video.

The full-project video must show compile, four distinct hosts, six peers, setup n=5/year 1996, the four specified queries, leave/query, join/query, teardown, deregistration, and manager shutdown. It must be at most 15 minutes, include audio, and have no splicing/edits.

References

Arizona State University, CSE 434, Fall 2026, Socket Programming Project, §§1.1–1.2.5, 2, 3 and 4.2.

NOAA/NCEI Storm Events Database: https://www.ncei.noaa.gov/pub/data/swdi/stormevents/csvfiles/